

Unified Mathematical Model — SEKHEM-TSDD

◆ 1) Global State Definition

$$\mathcal{S}_k = (\Psi_k, P_k, X_k, D_k, \gamma_k, u_k)$$

where:

- Ψ_k : system state
- P_k : probabilistic belief
- $X_k = (R_k, C_k, P_k)$: triadic state
- D_k : decision score
- $\gamma_k \in \{0,1\}$: validation gate
- u_k : control input

◆ 2) Core Unified Dynamics

$$\boxed{\mathcal{S}_{k+1} = F_{\{\mathrm{global}\}}(\mathcal{S}_k)}$$

◆ 3) Expanded Closed-Form System

بدل composition abstract، نكتبها explicit:

(a) State Evolution

$$\begin{aligned} \Psi_{k+1} &= \\ \Psi_k &+ \\ dt &F(\Psi_k, P_k) \\ &+ \\ \gamma_k &B(u_k) \\ &+ \\ \sigma \sqrt{dt} &\xi_k \end{aligned}$$

(b) Probabilistic Update

$$\begin{aligned} P_{k+1} &= \\ \mathcal{P} &(\Psi_k, P_k) \end{aligned}$$

(c) Triadic Construction

$$X_k = (R_k, C_k, P_k)$$

(d) PAC Mapping

$$\begin{aligned} & \Phi_{\{\mathrm{PAC}\}}(X_k) \\ &= \\ & X_k + \eta(\bar{X}_k - X_k) \end{aligned}$$

(e) Evaluation Layer

$$\begin{aligned} & D_u(k) \\ &= \\ & \sum_i W_i^k, \tilde{G}_{\{i,u\}}^k \end{aligned}$$

(f) Decision Rule

$$\begin{aligned} & u_k \\ &= \\ & \arg\max_{\{u \in \mathcal{U}_{\text{safe}}\}} D_u(k) \end{aligned}$$

(g) Validation Gate

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\gamma_k =  
\begin{cases}  
1 & I_{\mathrm{eff}} \geq \theta_{\mathrm{ind}} \\  
0 & \text{otherwise} \\  
\end{cases}
```

◆ 4) Unified Stability Condition (KEY EQUATION)

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\boxed{  
\dot{\mathcal{L}}(\mathcal{S})  
\leq  
- \alpha \|\Psi\|^2  
+  
\beta \sigma^2  
}
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◆ 5) Unified Safety Constraint

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\boxed{
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$$\gamma_k = 0 \rightarrow u_k = 0$$

◆ 6) Unified Robustness Model

$$\boxed{\begin{aligned} & \dot{\mathcal{L}} \\ & \leq \\ & - \alpha \|\Psi\|^2 \\ & + \\ & \gamma \|d\|^2 \end{aligned}}$$

◆ 7) Unified Optimal Control

$$\boxed{\begin{aligned} & u^*_k \\ & = \\ & \arg\min_{u \in \mathcal{U}_{\text{safe}}} \\ & \big(E(X_k) + \lambda J(u) \big) \end{aligned}}$$

◆ 8) Unified Closed Loop (FULL SYSTEM)

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\boxed{
\begin{aligned}
\Psi_{k+1} &= \Psi_k + dt \cdot F(\Psi_k) + \gamma_k \\
&\quad B(u_k) + \text{noise} \\
P_{k+1} &= \mathcal{P}(\Psi_k) \\
X_k &= (R_k, C_k, P_k) \\
u_k &= \arg\max D_u(k) \\
\gamma_k &= \mathbf{1}_{I_{\text{eff}} \geq \theta_{\text{ind}}} \\
\end{aligned}
}

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MASTER EQUATION (Final Form)

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\boxed{
\Psi_{k+1}
=
\Psi_k
+
dt \cdot F(\Psi_k)
+
\mathbf{1}_{I_{\text{eff}} \geq \theta_{\text{ind}}}
\cdot
B\left(
\arg\max_u D_u(\Phi_{\text{PAC}}(X_k))
\right)
}

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$$+ \int_0^t \int_{\mathbb{R}^d} \sigma(s, \mathbf{x}) \, d\mathbf{W}(s, \mathbf{x})$$

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